

Technical Summary:

Generating with Confidence: Uncertainty Quantification for Black-box Large Language Models

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1 Research Problem and Motivation

Uncertainty Quantification (UQ) for Natural Language Generation (NLG) addresses the need to estimate reliability of Large Language Model (LLM) outputs and to support selective deployment decisions, such as rejecting low-reliability cases to improve accuracy on the retained subset. The target operating regime is black-box LLM usage, where only generated text is available and token-level probabilities or logits are inaccessible due to closed APIs or restricted interfaces. This constraint makes many white-box UQ methods inapplicable.

Classical UQ methods from classification or regression do not directly transfer to free-form generation because the output space is extremely large, and multiple distinct strings can express the same meaning. Reliable decision making therefore requires uncertainty estimation in a semantic space rather than purely lexical space. Existing NLG-UQ approaches often focus on input-level uncertainty but then use it as a proxy for the correctness of an individual generation, which is insufficient when multiple candidate generations exist for the same input. The paper motivates an explicit separation between uncertainty as an input-level property and confidence as a response-level property.

2 Related Work

Several relevant strands motivate the proposed formulation.

Selective classification with rejection. Foundational work on recognition with rejection formalizes the tradeoff between accuracy and abstention and emphasizes a ranking score that identifies unreliable predictions. The paper maps selective NLG to this framework and argues that NLG requires response-dependent confidence scores in addition to input-dependent uncertainty.

UQ for NLG using semantic groupings. Recent work introduces semantic uncertainty and semantic entropy by grouping semantically equivalent generations and aggregating probability mass over semantic sets. These approaches typically require white-box access to generation probabilities, which conflicts with black-box LLM APIs.

Self-reported correctness. A separate baseline family asks a model to estimate whether a response is correct, commonly denoted as $P(\text{true})$. The paper treats this as computationally expensive when many samples are generated and uses it as a white-box baseline for non-API LLMs.

Lexical similarity baselines. Prior baselines for black-box settings often compute lexical overlap among multiple sampled generations. These methods ignore semantic equivalence and can be brittle to paraphrasing, motivating semantic similarity modeling.

Table 1: Datasets and splits used for evaluation.

Dataset	Task characteristics	Split used	Size	Notes
CoQA	Open-book conversational QA	Development	7,983	Evaluation only.
Natural Questions	Closed-book QA	Validation	3,610	Evaluation only.
TriviaQA (rc.nocontext)	Closed-book QA	Validation	9,960	De-duplicated questions.

3 Dataset Construction

Question answering (QA) datasets are used to enable scalable correctness evaluation based on reference answers rather than relying on expensive human evaluation for fully open-ended NLG. Three datasets are used:

- CoQA: an open-book conversational QA dataset.
- TriviaQA: a closed-book QA dataset (rc.nocontext subset) with de-duplicated questions.
- Natural Questions: a closed-book QA dataset described as more challenging.

Correctness labels are produced by an automatic judge model that assigns a score in $[0, 1]$ given the question, reference answer, and candidate response. A threshold of 0.7 is used to convert the score into a binary correctness label. Human verification of the auto-judgment is reported with accuracy about 0.95.

4 Query Protocol and Task Definitions

Generation protocol

Let x denote an input question and let an LLM produce a free-form response s . For each x , the black-box LLM is sampled m times to obtain m responses $\{s_1, \dots, s_m\}$. The default setting uses $m = 20$, and additional experiments vary m .

Uncertainty and confidence

Uncertainty is defined as an input-level property that reflects dispersion of the predictive distribution over responses. Confidence is defined as a response-level property that reflects reliability of a specific generated response for a given input. The paper evaluates both quantities under distinct protocols.

Evaluation settings and success criteria

Two task definitions are used.

Uncertainty + Expected Accuracy (U+EA). Each input x_i has an uncertainty score $U(x_i)$. The target is the expected correctness of the LLM on x_i , estimated by averaging correctness across m sampled generations:

$$\widehat{\mathbb{E}}[\text{acc}_i] = \frac{1}{m} \sum_{j=1}^m \text{acc}_{i,j}. \quad (1)$$

Purpose. Defines the evaluation target for input-level uncertainty ranking in the U+EA setting.

Symbols. m is the number of sampled generations; $\text{acc}_{i,j} \in \{0, 1\}$ is the binary correctness label for the

j -th generation of input x_i .

Intuition. Averaging across samples approximates how likely the LLM is to answer the input correctly under the same prompting and decoding configuration.

Usage. $U(x_i)$ is used to rank inputs for rejection; $\mathbb{E}[\text{acc}_i]$ supplies the accuracy values used to compute the accuracy-rejection curve.

Confidence + Indicator Accuracy (C+IA). Each individual generation $s_{i,j}$ is assigned a confidence score $C(x_i, s_{i,j})$. The target is the binary correctness indicator $\text{acc}_{i,j}$. Success is measured by ranking performance for correctness prediction.

Metrics

Two metrics are used.

AUROC (Area Under the Receiver Operating Characteristic curve). AUROC measures how well a score ranks correct generations above incorrect ones.

AUARC (Area Under the Accuracy-Rejection Curve). AUARC measures the area under the curve of retained accuracy versus rejection rate when samples are sorted by a score and the least reliable fraction is rejected. The oracle AUARC upper bound uses the target quantity itself as the score. The random baseline equals the base accuracy without rejection.

5 Modeling Approach

The approach is training-free and black-box compatible. It estimates uncertainty and confidence using only sampled text, through three steps:

1. Generate m responses for each input.
2. Compute pairwise similarity between responses in lexical or semantic space.
3. Aggregate similarity structure into an input-level uncertainty score and a response-level confidence score.

Similarity functions

Two similarity families are considered.

Jaccard similarity. Each response is treated as a set of words, and similarity is computed by intersection-over-union. This is efficient but ignores word order and can be sensitive to negation.

Natural Language Inference (NLI). An off-the-shelf NLI classifier provides probabilities for entailment and contradiction between two responses. These probabilities are used to define continuous similarity scores. This design remains compatible with black-box LLM usage because it does not require access to LLM logits.

Graph construction and spectral structure

For a fixed input, a weighted graph is formed over the m responses. Each response is a node, and edge weights are symmetrized pairwise similarities. Let W be the symmetric weighted adjacency matrix, and let D be the diagonal degree matrix. The symmetric normalized graph Laplacian is defined as:

$$L := I - D^{-\frac{1}{2}} W D^{-\frac{1}{2}}. \quad (2)$$

Purpose. Encodes cluster structure among sampled generations to support semantics-aware dispersion and consensus measures.

Symbols. I is the $m \times m$ identity matrix; $W \in \mathbb{R}^{m \times m}$ is the symmetric adjacency matrix with weights in $[0, 1]$; D is the degree matrix; m is the number of generations.

Intuition. Tight semantic agreement forms coherent clusters and yields a graph with strong internal connectivity, while semantic disagreement yields a diffuse graph.

Usage. Used as the spectral object for measures such as eigengap-based dispersion and spectral embeddings.

The degree matrix is:

$$D_{j_1, j_2} = \begin{cases} \sum_{j' \in [m]} w_{j_1, j'} & (j_1 = j_2), \\ 0 & (j_1 \neq j_2). \end{cases} \quad (3)$$

Purpose. Summarizes how strongly each response connects to others under the selected similarity function.

Symbols. $D \in \mathbb{R}^{m \times m}$; $D_{j,j}$ is the weighted degree of response j ; $w_{j_1, j'}$ is a symmetrized similarity weight.

Intuition. A response that agrees with many other samples has a high degree and lies near the consensus region.

Usage. Used directly to define degree-based uncertainty and confidence.

Degree-based uncertainty and confidence

The paper defines degree-based uncertainty and confidence as:

$$U_{\text{Deg}}(x) = \frac{\text{trace}(mI - D)}{m^2}, \quad C_{\text{Deg}}(x, s_j) = \frac{D_{j,j}}{m}. \quad (4)$$

Purpose. Provides a simple black-box uncertainty estimate for inputs and a confidence score for individual sampled generations.

Symbols. $\text{trace}(\cdot)$ is the matrix trace; D is the degree matrix in Eq. (3); m is the number of sampled generations; s_j is the j -th response for input x .

Intuition. When responses agree semantically, degrees increase and uncertainty decreases. A response is confident when it is well supported by other samples, reflected by a high degree.

Usage. $U_{\text{Deg}}(x)$ ranks inputs for rejection in AUARC evaluation; $C_{\text{Deg}}(x, s_j)$ ranks individual generations for AUROC and response-level selection.

Minimal end-to-end example

Given an input question, sample three responses ($m = 3$). If two responses are paraphrases that entail each other and the third expresses a different claim, the similarity graph contains a strong edge between the paraphrase pair and weaker edges to the outlier. The degree-based confidence assigns higher confidence to the paraphrase pair. The input-level uncertainty is lower than in a case where all three responses disagree semantically.

Experimental configuration and missing details

6 Empirical Results

Key results with required numbers

Table 3 summarizes representative AUARC results for U+EA. Table 4 summarizes representative AUARC results for C+IA. Higher AUARC indicates that ranking by the proposed score leads to larger accuracy gains as rejection increases, relative to random ranking and closer to the oracle upper bound.

Table 2: Models, components, and reported experimental details.

Item	Details
LLMs	OPT-13B, LLaMA-13B, LLaMA2, gpt-3.5-turbo (0301).
Generations per input	Default $m = 20$; additional experiments vary m (values reported in the paper).
Decoding parameters	Not specified in the paper.
Similarity models	Jaccard similarity; NLI similarity using an off-the-shelf DeBERTa-large NLI classifier with softmax probabilities.
Training	Not specified in the paper for base LLMs; the proposed method is training-free.
Baselines	NumSet, LexiSim, Semantic Entropy (white-box), $P(\text{true})$ (white-box).
Calibration	10 runs; each run uses 1,000 questions as a calibration set for hyperparameters and evaluates on the remainder.
Correctness labeling	Automatic judge with threshold 0.7; human verification accuracy about 0.95.
Compute and cost	Not specified in the paper.

Table 3: AUARC for Uncertainty + Expected Accuracy (U+EA) with $m = 3$ on TriviaQA (selected columns).

Method	trivia(LLaMA)	trivia(LLaMA2)	trivia(OPT)
Oracle	87.00 ± 0.05	90.45 ± 0.04	89.55 ± 0.05
Random	61.13 ± 0.06	76.20 ± 0.08	62.05 ± 0.07
LexiSim	77.79 ± 0.08	85.09 ± 0.07	82.53 ± 0.09
Deg (entailment similarity)	79.86 ± 0.08	85.83 ± 0.07	85.23 ± 0.08
Ecc (entailment similarity)	79.43 ± 0.09	85.90 ± 0.07	85.13 ± 0.08
Semantic Entropy (white-box)	77.53 ± 0.08	85.23 ± 0.08	84.47 ± 0.08
$P(\text{true})$ (white-box)	64.32 ± 0.15	84.61 ± 0.07	66.56 ± 0.18

Technical interpretation (U+EA). On trivia(LLaMA), Deg with entailment similarity increases AUARC from 61.13 (random) to 79.86, moving toward the oracle 87.00. This indicates that semantic dispersion among sampled generations is a strong predictor of expected correctness, enabling effective rejection. The $P(\text{true})$ baseline is substantially weaker on trivia(LLaMA) at 64.32, indicating that self-reported correctness under this configuration provides limited ranking value compared to semantic-consensus-based scoring.

Technical interpretation (C+IA). On trivia(LLaMA), Deg with entailment similarity achieves 85.27, exceeding LexiSim at 80.32 and approaching oracle at 87.06. This indicates that response-level consensus structure among samples predicts correctness of individual generations. Similar improvements hold for trivia(OPT), where Deg reaches 88.34 versus LexiSim at 83.64.

Ablation-style observations reported by the paper

The paper reports that the choice of similarity function has a larger impact than the choice among graph-based aggregation schemes. Entailment-based similarity outperforms contradiction-based similarity because low contradiction does not imply semantic agreement, and sets of low-quality responses can appear non-contradictory while still being unreliable.

The paper also reports an operational selective-generation effect: on trivia(LLaMA), rejecting the most uncertain 50% of questions using entailment-based degree scoring increases retained accuracy from about 62% to around 90%.

Table 4: AUARC for Confidence + Indicator Accuracy (C+IA) with $m = 20$ on TriviaQA (selected columns).

Method	trivia(LLaMA)	trivia(LLaMA2)	trivia(OPT)
Oracle	87.06 ± 0.04	90.84 ± 0.04	89.90 ± 0.05
Random	61.18 ± 0.07	76.24 ± 0.08	62.09 ± 0.07
LexiSim	80.32 ± 0.08	86.41 ± 0.06	83.64 ± 0.08
Deg (entailment similarity)	85.27 ± 0.07	88.87 ± 0.05	88.34 ± 0.06
Ecc (entailment similarity)	84.85 ± 0.07	89.05 ± 0.06	88.26 ± 0.06
Semantic Entropy (white-box)	79.15 ± 0.08	87.05 ± 0.06	87.13 ± 0.07
$P(\text{true})$ (white-box)	64.38 ± 0.16	84.65 ± 0.07	66.55 ± 0.18

Strengths and weaknesses grounded in results

Strengths. The proposed black-box methods using entailment similarity consistently outperform lexical baselines on the selected results. Degree-based confidence provides strong response-level ranking that is close to oracle AUARC in several settings, supporting selective generation and response selection among multiple samples.

Weaknesses. The evaluation depends on sampling diversity from the LLM. The paper states that when sampling temperature is zero and decoding becomes deterministic, the approach degenerates because repeated generations collapse to a single output, eliminating dispersion signals. Correctness labels rely on an automatic judge, and the evaluation is restricted to QA due to the difficulty of defining and collecting reliability labels for fully open-ended generation.

7 Summary

The paper formulates black-box UQ for NLG by explicitly separating input-level uncertainty from response-level confidence and by estimating both using only sampled generations. The core mechanism constructs a similarity graph over multiple sampled responses and derives dispersion and consensus measures, with entailment-based semantic similarity playing a central role. Quantitative results show that entailment-based degree scoring improves AUARC substantially over random and lexical baselines, and supports large gains in retained accuracy under rejection, including an example where retaining the most confident 50% raises accuracy from about 62% to around 90% on TriviaQA with LLaMA.

The paper states the following limitations and future directions. Evaluation is restricted to QA because open-ended NLG lacks clear reliability labels. Similarity choice is a primary dependency, and semantic equivalence is not guaranteed to be transitive, motivating continuous graph-based measures rather than hard clustering alone. The method depends on stochastic sampling and degenerates under deterministic decoding. Correctness evaluation relies on an automatic judge, despite reported human verification.